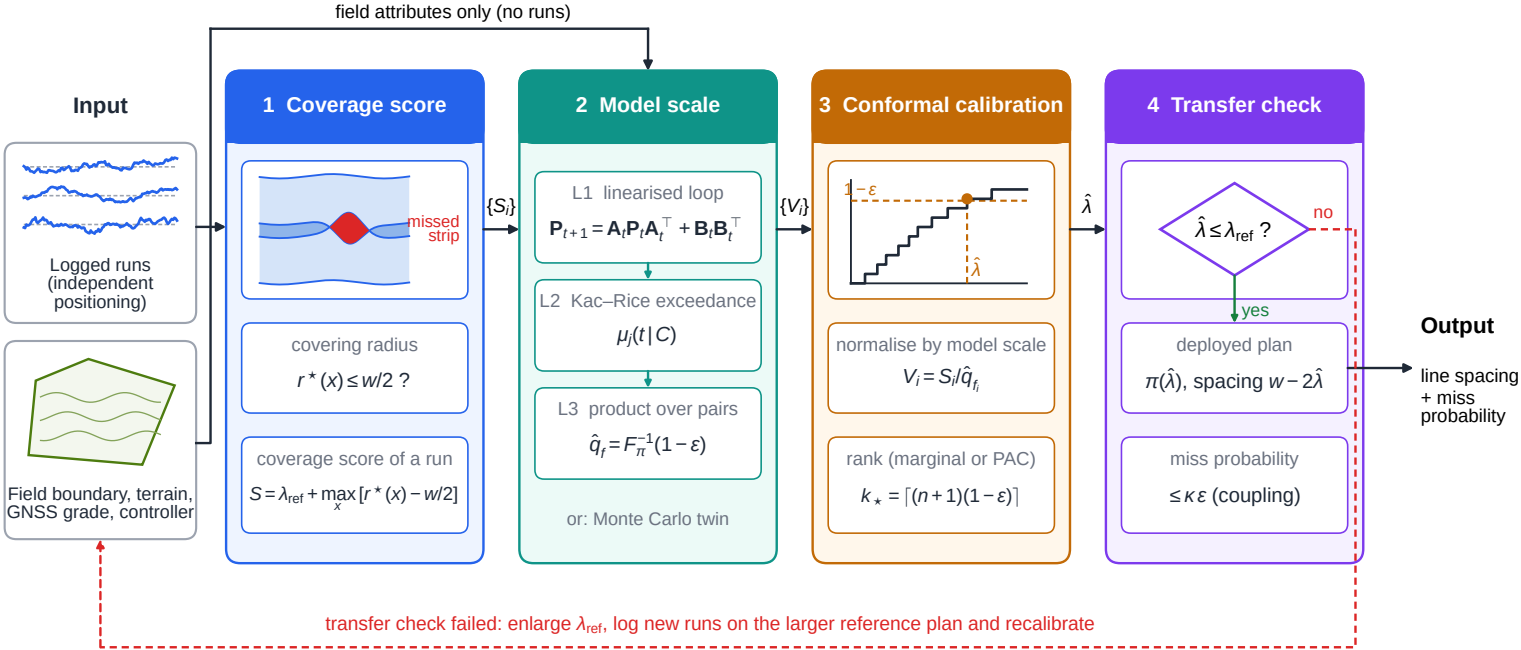
field attributes only (no runs)
Input
Logged runs (independent positioning)
Field boundary, terrain, GNSS grade, controller
1 Coverage score
missed strip
covering radius
$r^\star(x) \le w/2$?
coverage score of a run
$S = \lambda_{\text{ref}} + \max_x [r^\star(x) - w/2]$
$\{S_i\}$
2 Model scale
L1 linearised loop
$\mathbf{P}_{t+1} = \mathbf{A}_t \mathbf{P}_t \mathbf{A}_t^\top + \mathbf{B}_t \mathbf{B}_t^\top$
L2 Kac–Rice exceedance
$\mu_j(t \mid C)$
L3 product over pairs
$\hat{q}_f = F_\pi^{-1}(1-\varepsilon)$
or: Monte Carlo twin
$\{V_i\}$
3 Conformal calibration
$1-\varepsilon$
$\hat{\lambda}$
normalise by model scale
$V_i = S_i / \hat{q}_{f_i}$
rank (marginal or PAC)
$k_\star = \lceil (n+1)(1-\varepsilon) \rceil$
$\hat{\lambda}$
4 Transfer check
$\hat{\lambda} \le \lambda_{\text{ref}}$?
no
yes
deployed plan
$\pi(\hat{\lambda})$, spacing $w - 2\hat{\lambda}$
miss probability
$\le \kappa\varepsilon$ (coupling)
Output
line spacing + miss probability
transfer check failed: enlarge λ_{ref}, log new runs on the larger reference plan and recalibrate